

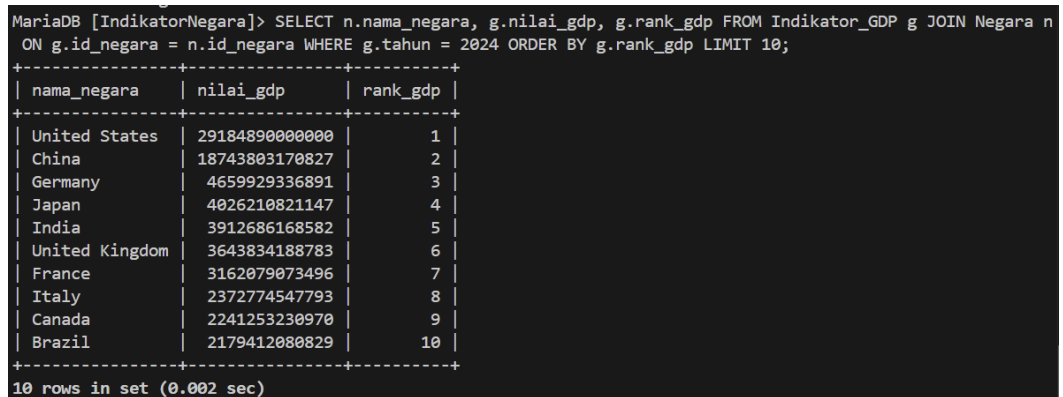
QUERY Data Storing

1. Top 10 Negara dengan GDP tertinggi tahun 2024

QUERY:

```
SELECT n.nama_negara, g.nilai_gdp, g.rank_gdp
FROM Indikator_GDP g
JOIN Negara n ON g.id_negara = n.id_negara
WHERE g.tahun = 2024
ORDER BY g.rank_gdp
LIMIT 10;
```

SCREENSHOT:



```
MariaDB [IndikatorNegara]> SELECT n.nama_negara, g.nilai_gdp, g.rank_gdp FROM Indikator_GDP g JOIN Negara n
ON g.id_negara = n.id_negara WHERE g.tahun = 2024 ORDER BY g.rank_gdp LIMIT 10;
+-----+-----+-----+
| nama_negara | nilai_gdp | rank_gdp |
+-----+-----+-----+
| United States | 2918489000000 | 1 |
| China | 18743803170827 | 2 |
| Germany | 4659929336891 | 3 |
| Japan | 4026210821147 | 4 |
| India | 3912686168582 | 5 |
| United Kingdom | 3643834188783 | 6 |
| France | 3162079073496 | 7 |
| Italy | 2372774547793 | 8 |
| Canada | 2241253230970 | 9 |
| Brazil | 2179412080829 | 10 |
+-----+-----+-----+
10 rows in set (0.002 sec)
```

2. Negara di Benua Asia dengan populasi > 100 juta dimulai dari negara dengan populasi terbesar

QUERY:

```
SELECT n.nama_negara, p.jumlah_populasi, b.nama_benua
FROM Negara n
JOIN Indikator_Populasi p ON n.id_negara = p.id_negara
JOIN Benua b ON n.id_benua = b.id_benua
WHERE b.nama_benua = "Asia" AND p.jumlah_populasi > 100000000
ORDER BY p.jumlah_populasi DESC;
```

SCREENSHOT:

```
MariaDB [IndikatorNegara]> SELECT n.nama_negara, p.jumlah_populasi, b.nama_benua FROM Negara n JOIN Indikator_Populasi p ON n.id_negara = p.id_negara JOIN Benua b ON n.id_benua = b.id_benua WHERE b.nama_benua = "Asia" AND p.jumlah_populasi > 100000000 ORDER BY p.jumlah_populasi DESC;
```

nama_negara	jumlah_populasi	nama_benua
India	1450935791	Asia
China	1419321278	Asia
Indonesia	283487931	Asia
Pakistan	251269164	Asia
Bangladesh	173562364	Asia
Japan	123753041	Asia
Philippines	115843670	Asia
Vietnam	100987686	Asia

8 rows in set (0.002 sec)

QUERY:

```
SELECT n.nama_negara, p.jumlah_populasi, b.nama_benua
FROM Negara n
JOIN Indikator_Populasi p ON n.id_negara = p.id_negara
JOIN Benua b ON n.id_benua = b.id_benua
WHERE p.jumlah_populasi > 100000000
AND EXISTS (
  SELECT 1 FROM Benua be WHERE be.id_benua = n.id_benua AND
  be.nama_benua = "Asia"
)
ORDER BY p.jumlah_populasi DESC;
```

SCREENSHOT:

```
MariaDB [IndikatorNegara]> SELECT n.nama_negara, p.jumlah_populasi, b.nama_benua FROM Negara n JOIN Indikator_Populasi p ON n.id_negara = p.id_negara JOIN Benua b ON n.id_benua = b.id_benua WHERE p.jumlah_populasi > 100000000 AND EXISTS ( SELECT 1 FROM Benua be WHERE be.id_benua = n.id_benua AND be.nama_benua = "Asia" ) ORDER BY p.jumlah_populasi DESC;
```

nama_negara	jumlah_populasi	nama_benua
India	1450935791	Asia
China	1419321278	Asia
Indonesia	283487931	Asia
Pakistan	251269164	Asia
Bangladesh	173562364	Asia
Japan	123753041	Asia
Philippines	115843670	Asia
Vietnam	100987686	Asia

8 rows in set (0.003 sec)

3. Indikator lengkap untuk Negara Indonesia (*just the important ones*)

QUERY:

```
SELECT n.nama_negara, b.nama_benua, g.nilai_gdp, g.rank_gdp,
p.jumlah_populasi, p.rank_populasi, c.emisi_co2, c.rank_co2,
e.konsumsi_energi_total, e.rank_energi, g.tahun
FROM Negara n
JOIN Benua b ON n.id_benua = b.id_benua
```

```

LEFT JOIN Indikator_GDP g ON n.id_negara = g.id_negara
LEFT JOIN Indikator_Populasi p ON n.id_negara = p.id_negara AND
p.tahun = g.tahun
LEFT JOIN Indikator_CO2 c ON n.id_negara = c.id_negara AND c.tahun
= g.tahun
LEFT JOIN Indikator_Energi e ON n.id_negara = e.id_negara AND
e.tahun = g.tahun
WHERE n.nama_negara = 'Indonesia';

```

SCREENSHOT:

```

MariaDB [IndikatorNegara]> SELECT n.nama_negara, b.nama_benua, g.nilai_gdp, g.rank_gdp, p.jumlah_populasi, p.rank_populasi, c.emisi_co2, c.rank_co2, e.konsum
si_energi_total, e.rank_energi, g.tahun FROM Negara n JOIN Benua b ON n.id_benua = b.id_benua LEFT JOIN Indikator_GDP g ON n.id_negara = g.id_negara LEFT JOI
n Indikator_Populasi p ON n.id_negara = p.id_negara AND p.tahun = g.tahun LEFT JOIN Indikator_CO2 c ON n.id_negara = c.id_negara AND c.tahun = g.tahun LEFT JOI
n Indikator_Energi e ON n.id_negara = e.id_negara AND e.tahun = g.tahun WHERE n.nama_negara = 'Indonesia';
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near '
n JOIN Benua b ON n.id_benua = b.id_benua LEFT JOIN Indikator_GDP g ON n.id_neg...' at line 1
MariaDB [IndikatorNegara]> SELECT n.nama_negara, b.nama_benua, g.nilai_gdp, g.rank_gdp, p.jumlah_populasi, p.rank_populasi, c.emisi_co2, c.rank_co2, e.konsum
si_energi_total, e.rank_energi, g.tahun FROM Negara n JOIN Benua b ON n.id_benua = b.id_benua LEFT JOIN Indikator_GDP g ON n.id_negara = g.id_negara LEFT JOI
n Indikator_Populasi p ON n.id_negara = p.id_negara AND p.tahun = g.tahun LEFT JOIN Indikator_CO2 c ON n.id_negara = c.id_negara AND c.tahun = g.tahun LEFT JOI
n Indikator_Energi e ON n.id_negara = e.id_negara AND e.tahun = g.tahun WHERE n.nama_negara = 'Indonesia';
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| nama_negara | nama_benua | nilai_gdp | rank_gdp | jumlah_populasi | rank_populasi | emisi_co2 | rank_co2 | konsumsi_energi_total | rank_energi | tahun |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Indonesia  | Asia       | 1396300098191 | 16 | 283487931 | 4 | 812204159 | 7 | NULL | NULL | 2024 |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.003 sec)

```

-- bagian energi NULL karena data energi hanya ada untuk tahun 2017

```

MariaDB [IndikatorNegara]> SELECT konsumsi_energi_total, rank_energi, tahun FROM Indikator_Energi e JOIN Negara n ON n.id_negara = e.id_negara WHERE nama_neg
ara = "Indonesia";
+-----+-----+-----+
| konsumsi_energi_total | rank_energi | tahun |
+-----+-----+-----+
| 7158688937000 | 15 | 2017 |
+-----+-----+-----+
1 row in set (0.002 sec)

```

4. Mendapatkan data negara dengan nilai GDP yang cukup tinggi, tapi tidak masuk top 30 negara penyumbang emisi CO2 tertinggi

QUERY:

```

SELECT n.nama_negara, g.rank_gdp, g.nilai_gdp, c.rank_co2,
c.emisi_co2
FROM Negara n
JOIN Indikator_GDP g ON n.id_negara = g.id_negara AND g.tahun =
2024
JOIN Indikator_CO2 c ON n.id_negara = c.id_negara AND c.tahun =
2024
WHERE g.rank_gdp IN (SELECT rank_gdp FROM Indikator_GDP WHERE
tahun = 2024 AND rank_gdp <= 20)
AND NOT EXISTS (
    SELECT 1
    FROM Indikator_CO2 c2
    WHERE c2.id_negara = n.id_negara
    AND c2.tahun = 2024
    AND c2.rank_co2 <= 30
)
ORDER BY rank_gdp DESC;

```

SCREENSHOT:

```
MariaDB [IndikatorNegara]> SELECT n.nama_negara, g.rank_gdp, g.nilai_gdp, c.rank_co2, c.emisi_co2
-> FROM Negara n
-> JOIN Indikator_GDP g ON n.id_negara = g.id_negara AND g.tahun = 2024
-> JOIN Indikator_CO2 c ON n.id_negara = c.id_negara AND c.tahun = 2024
-> WHERE g.rank_gdp IN (SELECT rank_gdp FROM Indikator_GDP WHERE tahun = 2024 AND rank_gdp <= 20)
-> AND NOT EXISTS (
-> SELECT 1
-> FROM Indikator_CO2 c2
-> WHERE c2.id_negara = n.id_negara
-> AND c2.tahun = 2024
-> AND c2.rank_co2 <= 30
-> )
-> ORDER BY rank_gdp DESC;
+-----+-----+-----+-----+-----+
| nama_negara | rank_gdp | nilai_gdp | rank_co2 | emisi_co2 |
+-----+-----+-----+-----+-----+
| Switzerland | 20 | 936564198049 | 68 | 33732136 |
| Netherlands | 19 | 1227543925316 | 38 | 118790884 |
+-----+-----+-----+-----+-----+
2 rows in set (0.001 sec)
```

5. Mendapatkan data negara yang mengalami resesif (*gdp growth negatif*) dan *population growth* yang menyusut juga secara bersamaan

QUERY:

```
SELECT n.nama_negara, g.persentase_pertumbuhan_gdp AS gdp_growth,
p.persentase_perubahan_tahunan AS pop_growth, g.rank_gdp,
p.rank_populasi
FROM Negara n
NATURAL JOIN Indikator_GDP g
NATURAL JOIN Indikator_Populasi p
WHERE g.tahun = 2024 AND p.tahun = 2024 AND
g.persentase_pertumbuhan_gdp < 0 AND
p.persentase_perubahan_tahunan < 0
ORDER BY gdp_growth DESC;
```

SCREENSHOT:

```
MariaDB [IndikatorNegara]> SELECT n.nama_negara, g.persentase_pertumbuhan_gdp AS gdp_growth, p.persentase_perubahan_tahunan AS pop_growth, g
.rank_gdp, p.rank_populasi
-> FROM Negara n
-> NATURAL JOIN Indikator_GDP g
-> NATURAL JOIN Indikator_Populasi p
-> WHERE g.tahun = 2024 AND p.tahun = 2024 AND g.persentase_pertumbuhan_gdp < 0 AND p.persentase_perubahan_tahunan < 0
-> ORDER BY gdp_growth DESC;
+-----+-----+-----+-----+-----+
| nama_negara | gdp_growth | pop_growth | rank_gdp | rank_populasi |
+-----+-----+-----+-----+-----+
| Estonia | -0.26 | -0.49 | 105 | 157 |
| Latvia | -0.44 | -0.56 | 103 | 151 |
| Jamaica | -0.72 | -0.02 | 131 | 142 |
| Austria | -1.17 | -0.11 | 31 | 99 |
| U.S. Virgin Islands | -1.31 | -0.93 | 172 | 202 |
| Cuba | -1.93 | -0.36 | 57 | 90 |
| Isle of Man | -4.17 | -0.01 | 162 | 200 |
+-----+-----+-----+-----+-----+
7 rows in set (0.001 sec)
```

6. Mengecek check constraint untuk Indikator GDP

QUERY:

```
INSERT INTO Indikator_GDP VALUES (1, 2024, -900, 1.0, 100, 99, 2);  
INSERT INTO Indikator_GDP VALUES (1, 2024, 900, 1.0, -100, 99, 2);  
INSERT INTO Indikator_GDP VALUES (1, 2024, 900, 1.0, 100, 101, 2);
```

-- nilai_gdp & gdp_per_capita tidak boleh negatif, serta persentase_gdp_dunia between 0 and 100

SCREENSHOT:

```
MariaDB [IndikatorNegara]> INSERT INTO Indikator_GDP VALUES (1, 2024, -900, 1.0, 100, 99, 2);  
ERROR 4025 (23000): CONSTRAINT `Indikator_GDP.nilai_gdp` failed for `IndikatorNegara`.`Indikator_GDP`  
MariaDB [IndikatorNegara]> INSERT INTO Indikator_GDP VALUES (1, 2024, 900, 1.0, -100, 99, 2);  
ERROR 4025 (23000): CONSTRAINT `Indikator_GDP.gdp_per_capita` failed for `IndikatorNegara`.`Indikator_GDP`  
MariaDB [IndikatorNegara]> INSERT INTO Indikator_GDP VALUES (1, 2024, 900, 1.0, 100, 101, 2);  
ERROR 4025 (23000): CONSTRAINT `Indikator_GDP.persentase_gdp_dunia` failed for `IndikatorNegara`.`Indikator_GDP`
```

7. Mengecek trigger trg_indikator_before_komposisi_insert

QUERY:

```
INSERT INTO Komposisi_Energi VALUES (1, 2024, 1, 1000000, 50);
```

SCREENSHOT:

```
MariaDB [IndikatorNegara]> INSERT INTO Komposisi_Energi VALUES (1, 2024, 1, 1000000, 50);  
ERROR 1644 (45000): Data komposisi energi harus punya data indikator energi terlebih dahulu
```

8. Mengecek trigger trg_persentase_energi_insert

QUERY:

```
INSERT INTO Komposisi_Energi VALUES (1, 2017, 1, 100, 50);
```

SCREENSHOT:

```
MariaDB [IndikatorNegara]> INSERT INTO Komposisi_Energi VALUES (1, 2017, 1, 100, 50);  
ERROR 1644 (45000): Error Insert: Total persentase konsumsi energi tidak boleh melebihi 100%
```

9. Mengecek trigger trg_persentase_energi_dunia_update

QUERY:

```
UPDATE Indikator_Energi SET persentase_dunia = 20 WHERE id_negara
```

```
= 1;
```

SCREENSHOT:

```
MariaDB [IndikatorNegara]> UPDATE Indikator_Energi SET persentase_dunia = 20 WHERE id_negara = 1;  
ERROR 1644 (45000): Error Update: Total persentase konsumsi energi dunia tidak boleh melebihi 101%
```

10. Mengecek procedure insert_gdp, update_gdp, dan delete_gdp

QUERY:

```
CALL insert_gdp(230, 2024, 70000000, 3.0, 10000, 6.5e-05);
```

SCREENSHOT:

```
MariaDB [IndikatorNegara]> CALL insert_gdp(230, 2024, 70000000, 3.0, 10000, 6.5e-05);  
Query OK, 3 rows affected, 1 warning (0.044 sec)
```

-- mencari id_negara yang blum ada indikator dulu

```
MariaDB [IndikatorNegara]> SELECT n.id_negara FROM Negara n LEFT JOIN Indikator_GDP g ON n.id_negara = g.id_negara WHERE g.nilai_gdp IS NULL  
?  
+-----+  
| id_negara |  
+-----+  
| 132 |  
| 172 |  
| 174 |  
| 231 |  
| 80 |  
| 91 |  
| 38 |  
| 69 |  
| 73 |  
| 84 |  
| 129 |  
| 173 |  
| 178 |  
| 152 |  
| 211 |  
| 230 |  
+-----+  
16 rows in set (0.001 sec)
```

-- menyisipkan pada posisi rank_gdp 1 sebelum paling bawah (negara Tuvalu)

```
MariaDB [IndikatorNegara]> SELECT * FROM Indikator_GDP WHERE id_negara = 230;
+-----+-----+-----+-----+-----+-----+
| id_negara | tahun | nilai_gdp | persentase_pertumbuhan_gdp | gdp_per_capita | persentase_gdp_dunia | rank_gdp |
+-----+-----+-----+-----+-----+-----+
|      230 | 2024 | 70000000 |                3.00 |      10000 |                0.00 |      218 |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.001 sec)
```

```
MariaDB [IndikatorNegara]> SELECT nama_negara, rank_gdp FROM Indikator_GDP g JOIN Negara n ON n.id_negara = g.id_negara WHERE nama_negara = "Tuvalu";
+-----+-----+
| nama_negara | rank_gdp |
+-----+-----+
| Tuvalu      |      219 |
+-----+-----+
1 row in set (0.001 sec)
```

QUERY:

```
CALL update_gdp(230, 2024, 85000000, 4.0, 20000, 8e-05);
```

SCREENSHOT:

```
MariaDB [IndikatorNegara]> CALL update_gdp(230, 2024, 85000000, 4.0, 20000, 8e-05);
Query OK, 8 rows affected, 1 warning (0.017 sec)
```

-- rank negara dengan id 230 (Wallis & Futuna) naik ke 217

```
MariaDB [IndikatorNegara]> SELECT * FROM Indikator_GDP WHERE id_negara = 230;
+-----+-----+-----+-----+-----+-----+
| id_negara | tahun | nilai_gdp | persentase_pertumbuhan_gdp | gdp_per_capita | persentase_gdp_dunia | rank_gdp |
+-----+-----+-----+-----+-----+-----+
|      230 | 2024 | 85000000 |                4.00 |      20000 |                0.00 |      217 |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.001 sec)
```

QUERY:

```
CALL delete_gdp(230, 2024);
```

SCREENSHOT:

```
MariaDB [IndikatorNegara]> CALL delete_gdp(230, 2024);
Query OK, 3 rows affected (0.007 sec)
```

-- rank negara Tuvalu kembali ke 218

```
MariaDB [IndikatorNegara]> SELECT nama_negara FROM Indikator_GDP g NATURAL JOIN Negara n WHERE rank_gdp = 218;
+-----+
| nama_negara |
+-----+
| Tuvalu      |
+-----+
1 row in set (0.001 sec)
```

